

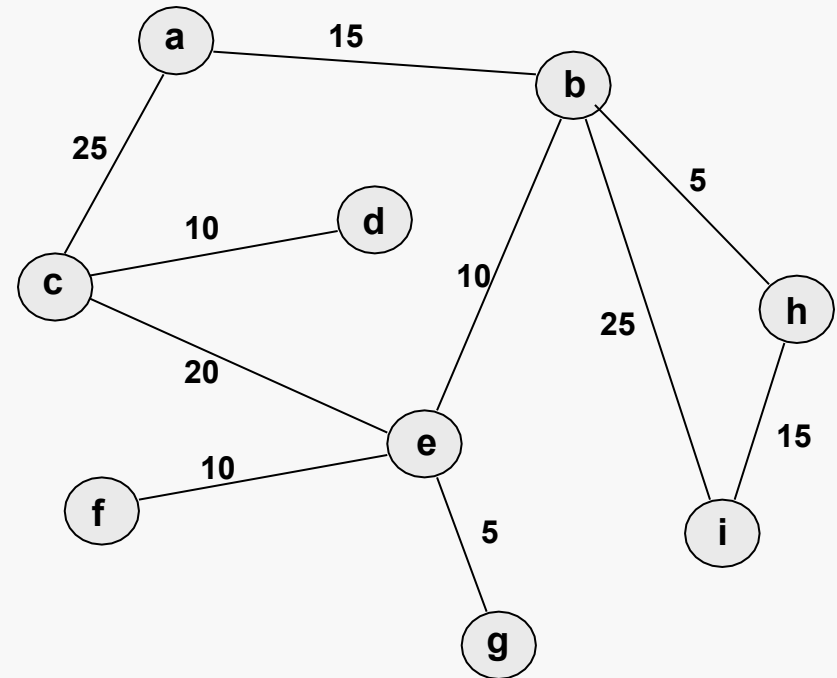
Weighted Graphs

Weighted Graphs 1

In many applications, each edge of a graph has an associated numerical value, called a weight.

Usually, the edge weights are non-negative integers.

Weighted graphs may be either directed or undirected.



The weight of an edge is often referred to as the "cost" of the edge.

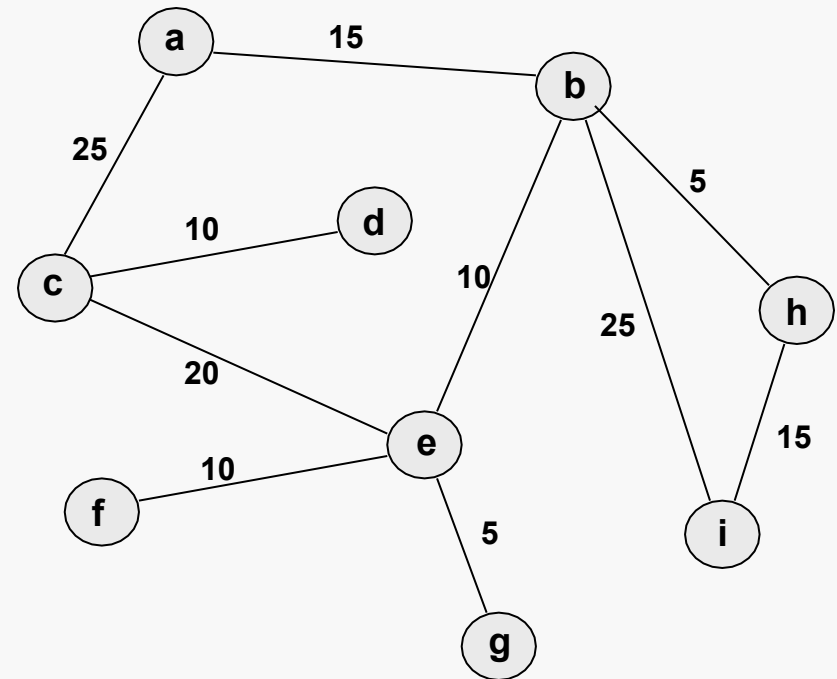
In applications, the weight may be a measure of the length of a route, the capacity of a line, the energy required to move between locations along a route, etc.

Shortest Paths (SSAD)

Weighted Graphs 2

Given a weighted graph, and a designated node S , we would like to find a path of least total weight from S to each of the other vertices in the graph.

The total weight of a path is the sum of the weights of its edges.



We have seen that performing a DFS or BFS on the graph will produce a spanning tree, but neither of those algorithms takes edge weights into account.

There is a simple, greedy algorithm that will solve this problem.

Dijkstra's SSAD Algorithm*

Weighted Graphs 3

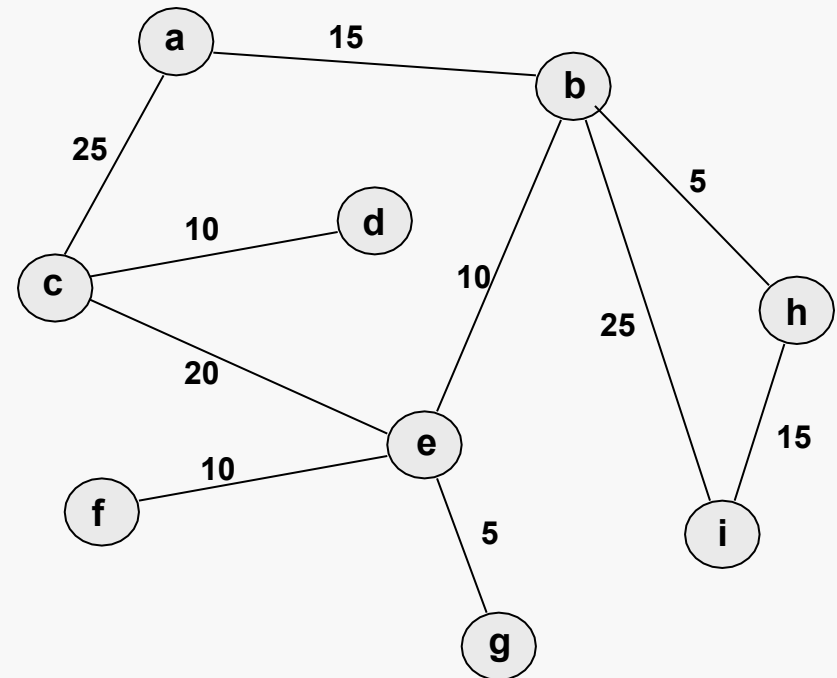
We assume that there is a path from the source vertex s to every other vertex in the graph.

Let S be the set of vertices whose minimum distance from the source vertex has been found. Initially S contains only the source vertex.

The algorithm is iterative, adding one vertex to S on each pass.

We maintain a table D such that for each vertex v , $D(v)$ is the minimum distance from the source vertex to v via vertices that are already in S (aside possibly from v itself).

Greed: on each iteration, add to S the vertex v not already in S for which $D(v)$ is minimal.

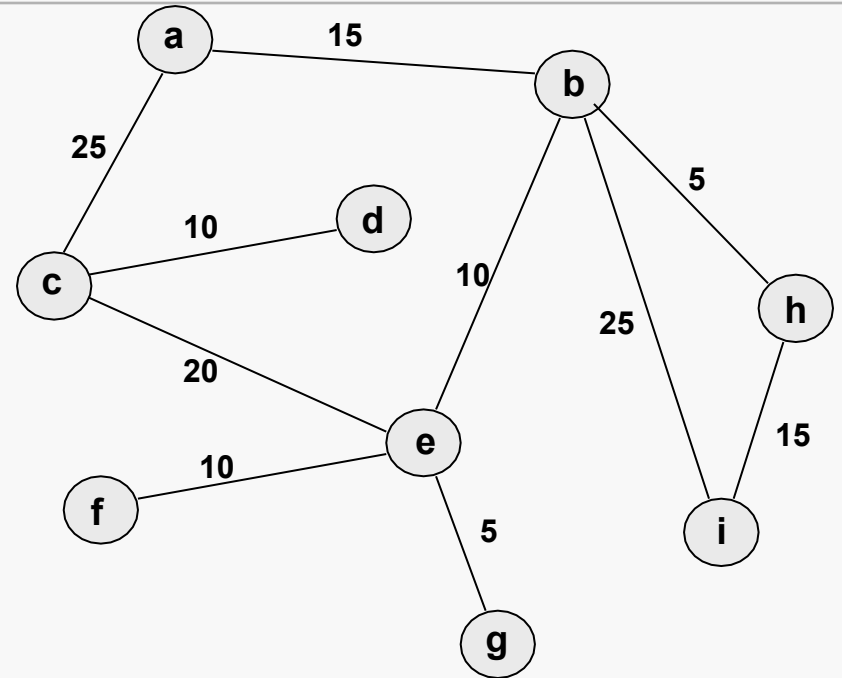


*1959

Dijkstra's Algorithm Trace

Weighted Graphs 4

Let the source vertex be a.



$S = \{a\}$

D

a	b	c	d	e	f	g	h	i
0	15	25	∞	∞	∞	∞	∞	∞
null	a	a	null	null	null	null	null	null

$S = \{a,b\}$

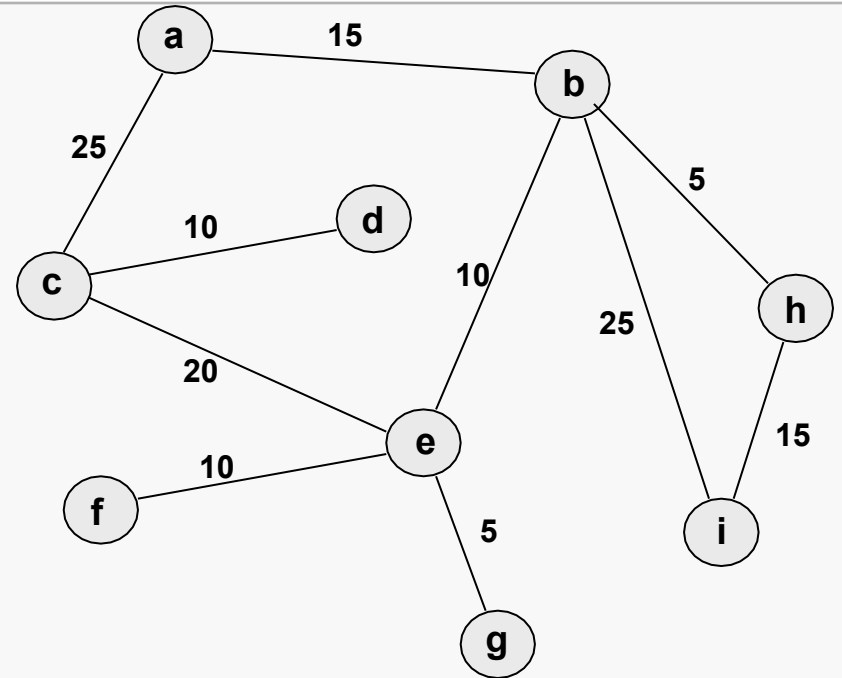
D

a	b	c	d	e	f	g	h	i
0	15	25	∞	25	∞	∞	20	40
null	a	a	null	b	null	null	b	b

Dijkstra's Algorithm Trace

Weighted Graphs 4

Let the source vertex be a.



$S = \{a, b, h\}$

D	a	b	c	d	e	f	g	h	i
	0	15	25	∞	25	∞	∞	20	35
	null	a	a	null	h	null	null	b	h

$S = \{a, b, h, c\}$

D	a	b	c	d	e	f	g	h	i
	0	15	25	35	25	∞	∞	20	35
	null	a	a	c	b	null	null	b	h

Dijkstra's Algorithm Trace

Weighted Graphs 4

$S = \{a, b, h, c, e\}$

D

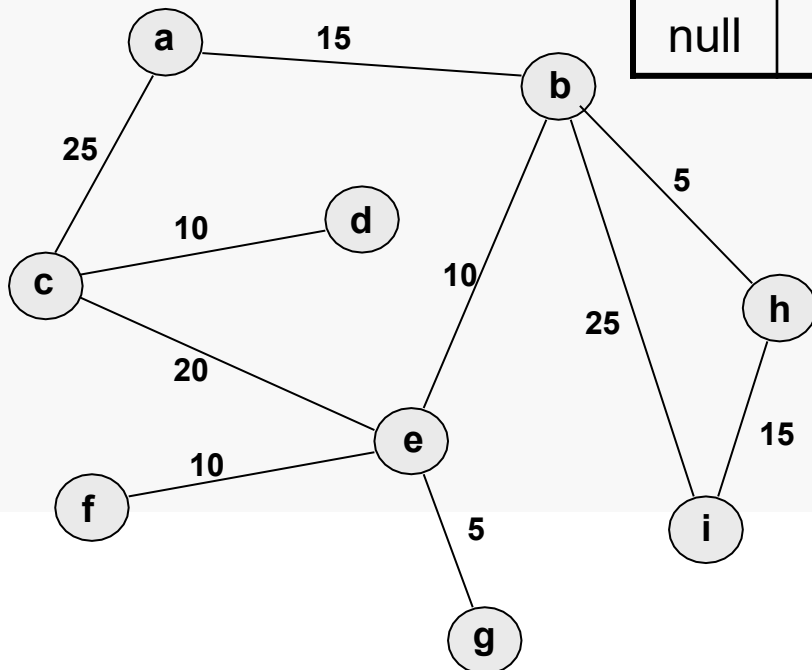
a	b	c	d	e	f	g	h	i
0	15	25	35	25	35	30	20	35
null	a	a	c	h	e	e	b	h

$S = \{a, b, h, c, e, g\}$

a	b	c	d	e	f	g	h	i
0	15	25	35	25	35	30	20	35
null	a	a	c	b	e	e	b	h

$S = \{a, b, h, c, e, g, f\}$

a	b	c	d	e	f	g	h	i
0	15	25	35	25	35	30	20	35
null	a	a	c	b	e	e	b	h



Which distances changed while processing nodes g and f?

Dijkstra's Algorithm Trace

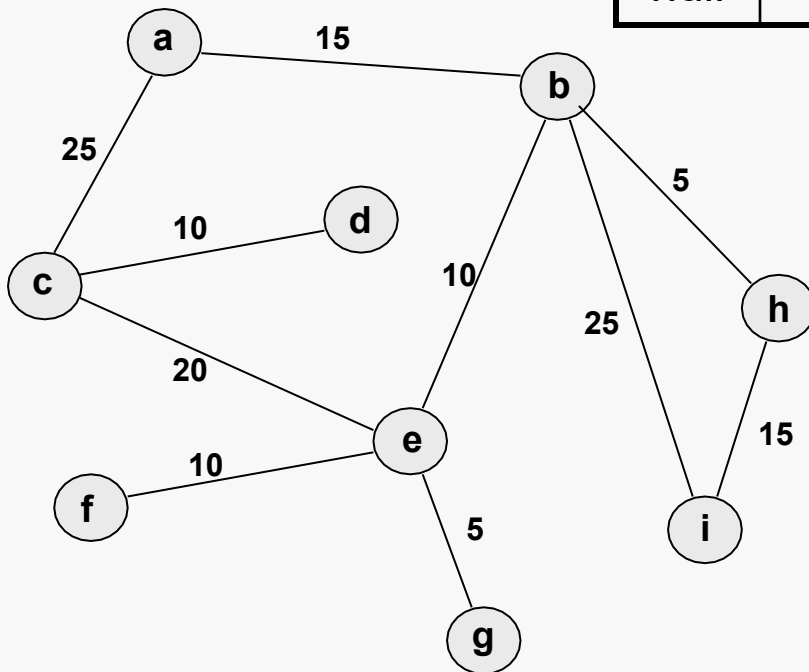
Weighted Graphs 6

$S = \{a, b, h, c, e, g, f, d\}$

a	b	c	d	e	f	g	h	i
0	15	25	35	25	35	30	20	35
null	a	a	c	b	e	e	b	h

$S = \{a, b, h, c, e, g, f, d, i\}$

a	b	c	d	e	f	g	h	i
0	15	25	35	25	35	30	20	35
null	a	a	c	b	e	e	b	h



What is the shortest path from **a** to **i**?

abhi

Dijkstra's SSAD Algorithm only works for graphs with non-negative weights.

See the Bellman-Ford Algorithm, which works even if the weights are negative, provided there is no *negative cycle* (a cycle whose total weight is negative).

The corresponding tree is shown and isn't the minimum spanning tree for this graph.

I.e., it doesn't have the least sum of weights possible. For that we need to use Prim's algorithm.

